Web Design Project 2

Group 6 :

Bagas Pratama Junianika

Naufal Fayyaz Ilham

Radella Hafiza Gehan

Saw Innocent Sine

Mohammad Naufal Pranantya

Introduction

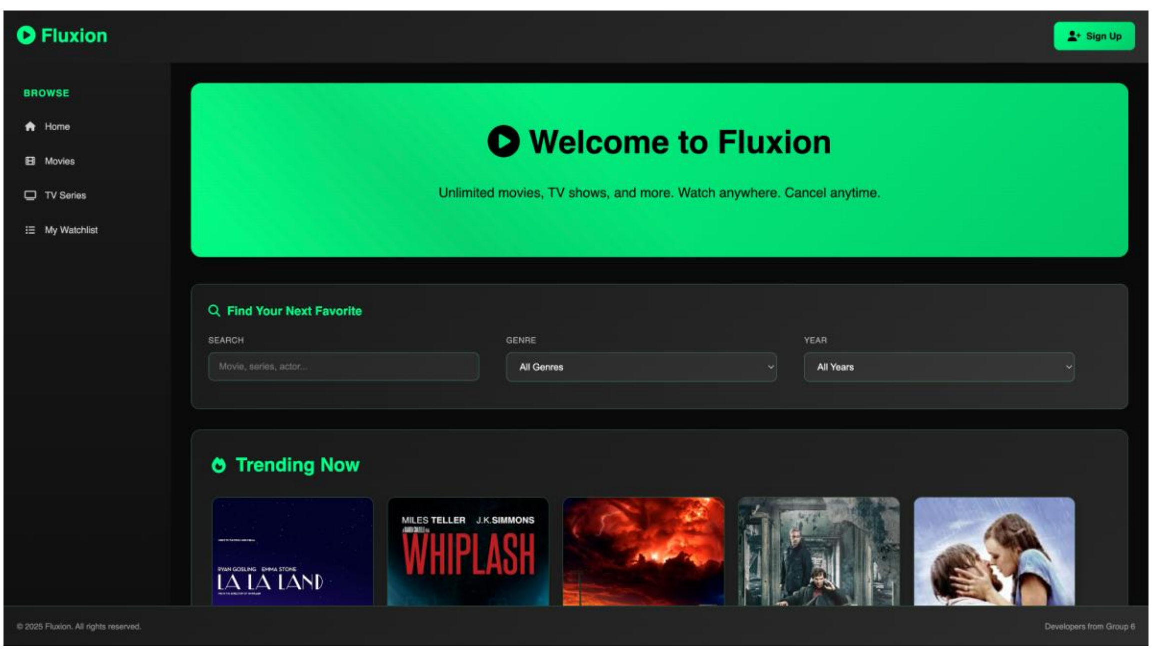
Fluxion is a Netflix-style movie and TV series streaming platform built as an educational web development project. The website allows users to browse, discover, and watch movies and series through a modern dark-themed interface with vibrant green accents.

The platform serves as an entertainment hub with a curated collection of movies and TV series. Users can search and filter content by genres, ratings, and other criteria. The website features a professional dark theme design with green accent colors (#00ff88) that creates an engaging visual experience.

As an educational project, Fluxion demonstrates mastery of HTML5, CSS3, and web development principles using traditional frameset architecture. The platform is fully responsive, working across desktop, tablet, and mobile devices. It includes complete HTML5 form validation, 10 individual watch pages with embedded video players, and a comprehensive sign-up system with multiple subscription tiers.

Page-by-page Analysis

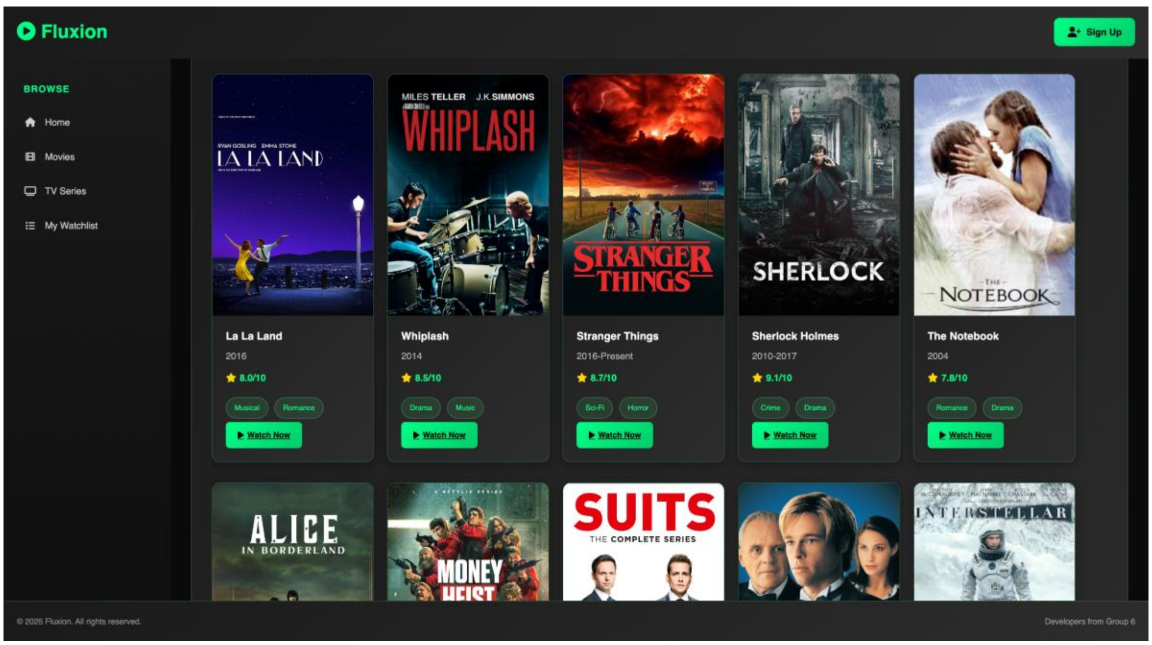
Master Frameset (index.html)



The master frameset is the main entry point that creates the website's four-section layout using traditional HTML frameset architecture. The layout includes a header frame (80px height) with branding and sign-up functionality, a navigation frame (250px width) with the sidebar menu, a content frame that loads different pages dynamically, and a footer frame (60px height) with copyright information.

Key features include internal CSS for frame styling, a noframes fallback for browser compatibility, automatic loading of home.html as the default page, and responsive design that adapts to different screen sizes. The technical implementation uses external CSS, Google Fonts (Poppins), Font Awesome icons, and gradient backgrounds for a professional appearance.

Home Page (home.html)

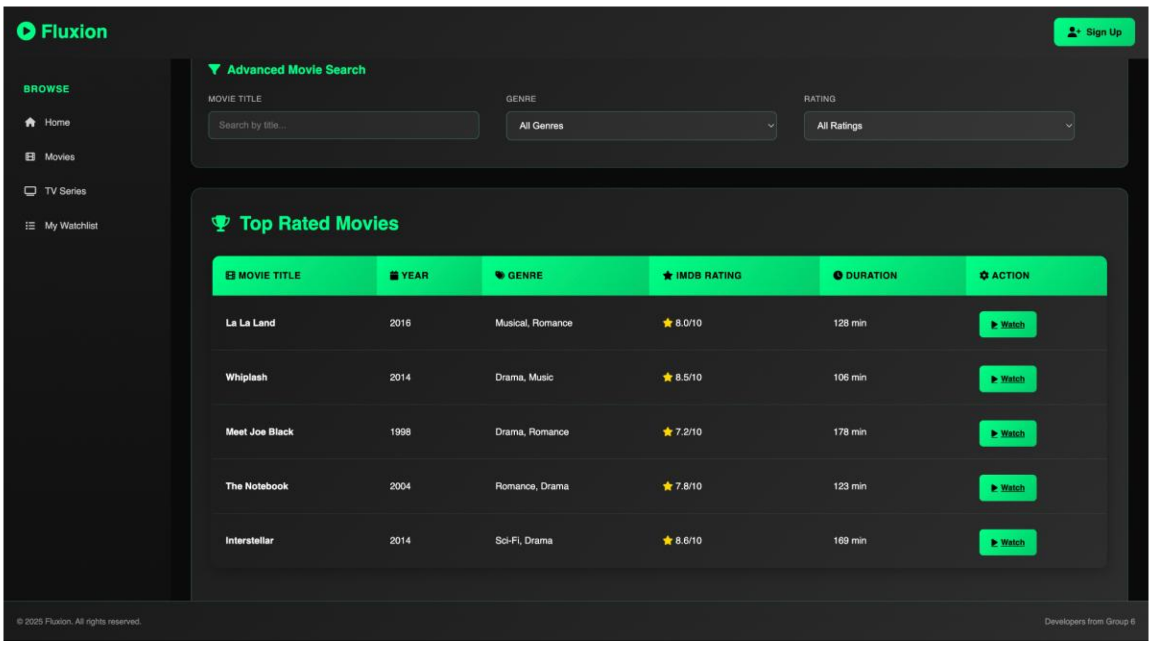


The home page is the main landing page and content discovery hub. It features a hero banner with animated shimmer effects and the tagline "Unlimited movies, TV shows, and more. Watch anywhere. Cancel anytime."

The search and filter system includes a text search field for movies, series, and actors, a genre filter with options (Action, Comedy, Drama, Horror, Sci-Fi, Thriller), and a year filter (2021-2025). The trending content grid displays 10 featured titles in a responsive layout, including La La Land (2016, Musical/Romance, 8.0/10), Whiplash (2014, Drama/Music, 8.5/10), Stranger Things (2016-Present, Sci-Fi/Horror, 8.7/10), Sherlock Holmes (2010-2017, Crime/Drama, 9.1/10), The Notebook (2004, Romance/Drama, 7.8/10), Alice in Borderland (2020-Present, Sci-Fi/Thriller, 7.6/10), Money Heist (2017-2021, Crime/Thriller, 8.3/10), The Suits (2011-2019, Drama/Comedy, 8.5/10), Meet Joe Black (1998, Drama/Romance, 7.2/10), and Interstellar (2014, Sci-Fi/Drama, 8.6/10).

Each content card shows movie posters, titles, years, IMDb ratings with stars, genre badges, and "Watch Now" buttons. The "Why Choose Fluxion" section highlights key features like unlimited streaming, multi-device access, HD/4K quality, and offline downloads in a two-column layout. Technical features include inline CSS with shimmer animations, responsive grid system, hover effects, and Font Awesome icons.

Movie Page (movies.html)



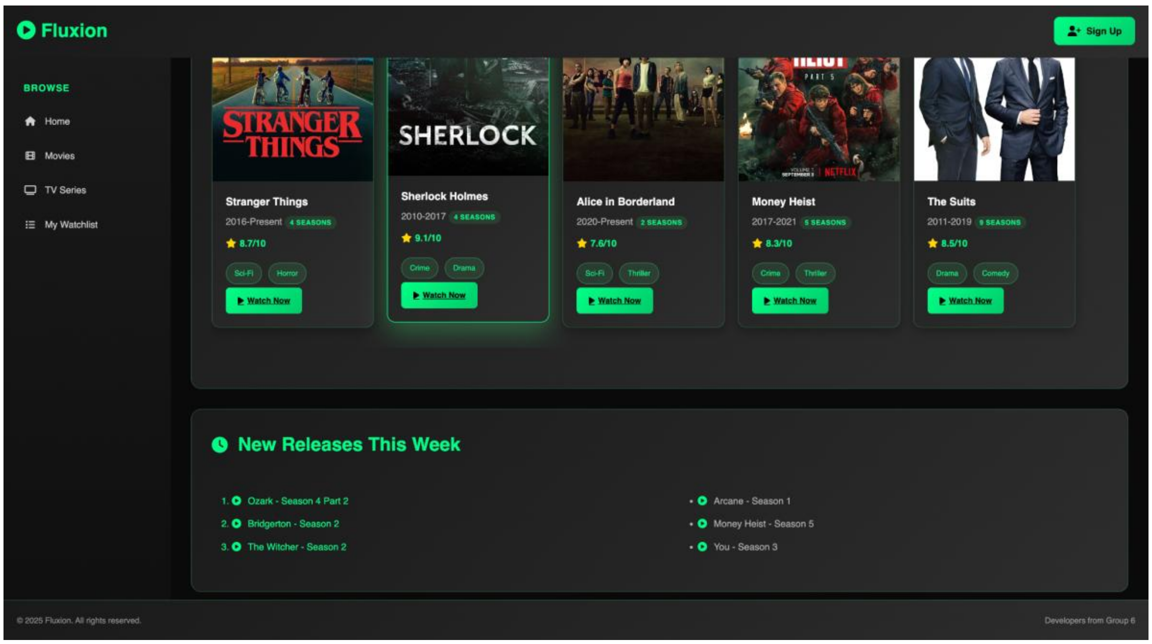
The movies page is dedicated to browsing and discovering movie content. It features a "Movies Collection" hero banner with film icon and gradient styling.

A statistics dashboard shows 5 movies, 25+ genres, 4K quality, and 24/7 availability. The advanced search includes movie title search, genre filter (Action & Adventure, Comedy, Drama, Horror, Sci-Fi & Fantasy, Thriller, Romance, Documentary), and rating filter (9.0+, 8.0+, 7.0+, 6.0+).

The top rated movies table displays five featured titles: La La Land (2016, Musical/Romance, 8.0/10, 128 min), Whiplash (2014, Drama/Music, 8.5/10, 106 min), Meet Joe Black (1998, Drama/Romance, 7.2/10, 178 min), The Notebook (2004, Romance/Drama, 7.8/10, 123 min), and Interstellar (2014, Sci-Fi/Drama, 8.6/10, 169 min). Each row includes movie title with film icon, year with calendar icon, genre tags, IMDb rating with stars, duration with clock icon, and "Watch" button.

The browse by genre section shows genre categories with movie counts in two columns: Action & Adventure (1,250), Comedy (980), Drama (1,500), Horror (650), Sci-Fi & Fantasy (420), Romance (750), Thriller (580), and Documentary (320). Technical features include inline CSS for statistics styling, professional table design with hover effects, responsive layout, and Font Awesome icons.

Series Page (series.html)

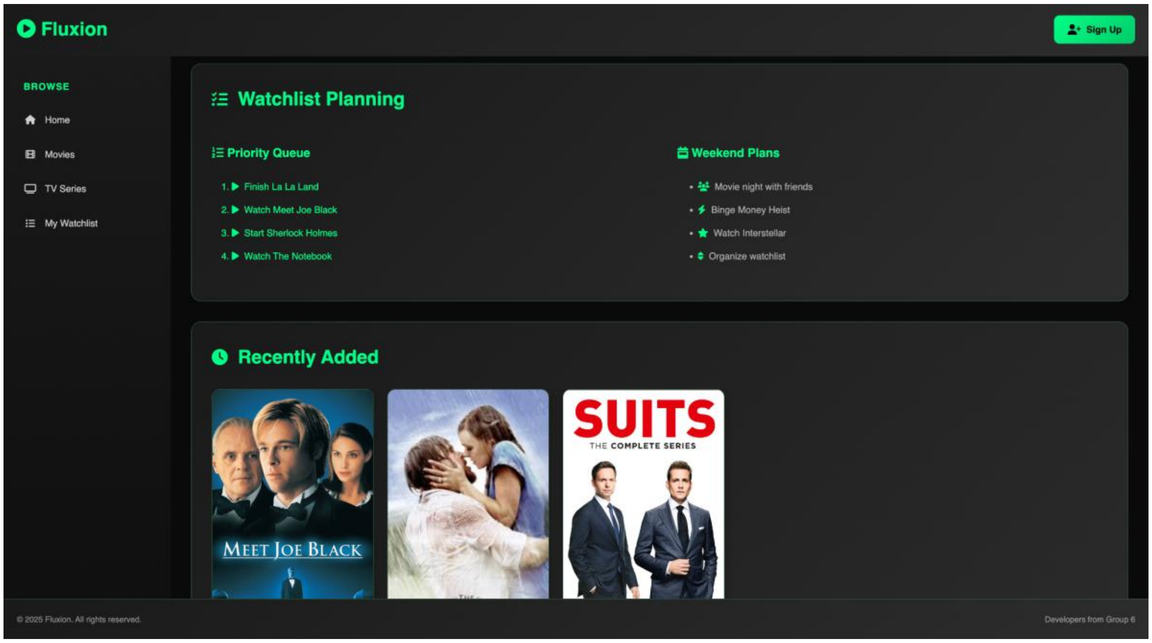


The series page focuses on TV series content with season details and status tracking. It features a "TV Series Collection" hero banner with TV icon and binge-watching emphasis.

The search interface includes series title search, genre filter (Crime, Drama, Comedy, Sci-Fi, Fantasy, Thriller, Horror), and status filter (Ongoing, Completed, New Releases).

The popular series grid displays five featured series: Stranger Things (2016-Present, 4 Seasons, Sci-Fi/Horror, 8.7/10, ongoing), Sherlock Holmes (2010-2017, 4 Seasons, Crime/Drama, 9.1/10, completed), Alice in Borderland (2020-Present, 2 Seasons, Sci-Fi/Thriller, 7.6/10, ongoing), Money Heist (2017-2021, 5 Seasons, Crime/Thriller, 8.3/10, completed), and The Suits (2011-2019, 9 Seasons, Drama/Comedy, 8.5/10, completed).

Playlist Page (playlist.html)



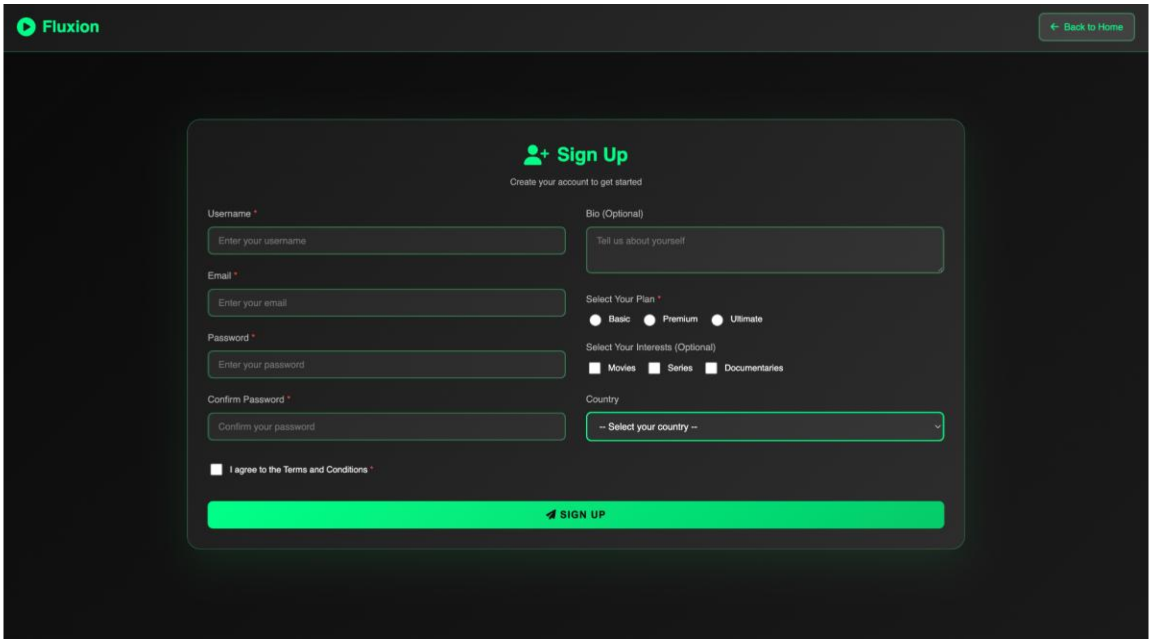
The playlist page is a personal watchlist management system for tracking viewing progress and planning future content. It features a "My Watchlist" hero banner with list icon and user-focused messaging.

A statistics dashboard shows 24 items in progress, 156 completed, 89 to watch, and 12 favorites. The currently watching section displays three items with progress bars: La La Land (45% progress, Musical/Romance, 8.0/10), Stranger Things (40% progress, Sci-Fi/Horror, 8.7/10), and Whiplash (30% progress, Drama/Music, 8.5/10).

The watchlist planning section has two columns: a priority queue (finish La La Land, watch Meet Joe Black, start Sherlock Holmes, watch The Notebook) and weekend plans (movie night with friends, binge Money Heist, watch Interstellar, organize watchlist).

The recently added section shows Meet Joe Black (1998, Drama/Romance, 7.2/10), The Notebook (2004, Romance/Drama, 7.8/10), and The Suits (2011-2019, Drama/Comedy, 8.5/10). Technical features include custom progress bar styling, statistics dashboard with gradient backgrounds, responsive grid layout, and interactive planning tools.

Sign Up Page (signup.html)



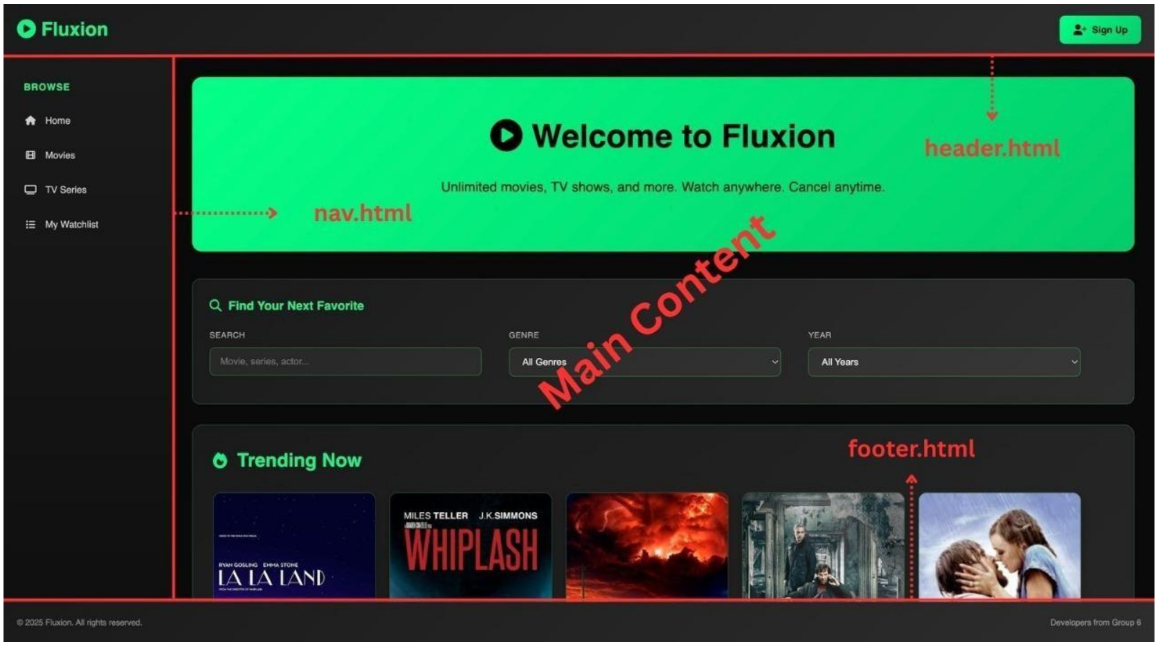
The sign-up page is a comprehensive user registration system with complete HTML5 form validation and modern UI design. It features a navigation bar with Fluxion logo and "Back to Home" button.

The registration form uses a two-column responsive layout. Column 1 includes username (3-20 chars, alphanumeric + underscore), email (standard validation), password (8+ chars, uppercase, lowercase, number), and confirm password (pattern matching). Column 2 includes bio (optional, 200 char limit), subscription plan (Basic, Premium, Ultimate radio buttons), interests (Movies, Series, Documentaries checkboxes), and country (dropdown selection).

Full-width elements include required terms and conditions checkbox and gradient-styled submit button with hover effects. Form validation features HTML5 native validation with no JavaScript dependencies, visual feedback (green/red borders), real-time validation, pattern matching, and required field indicators (red asterisks).

Technical features include responsive CSS Grid design, custom validation styling, smooth transitions and hover effects, professional form design, and mobile-optimized layout.

Navigation Components



Header Component (header.html)

The header component provides branding and user account access throughout the website. It includes a logo section with Fluxion branding and play circle icon, and a user section with Sign Up button and user-plus icon. Features include 80px fixed height, gradient background, hover effects, responsive design, and links to home page and sign-up form. Technical implementation uses internal CSS, flexbox layout, Font Awesome icons, and smooth transitions.

Navigation Component (nav.html)

The navigation component is the sidebar navigation system with main site navigation and filtering options. It includes a browse section with navigation items: Home (home icon), Movies (film icon), TV Series (TV icon), and My Watchlist (list icon). Features include 250px fixed width, gradient background, hover effects with slide animation, icon-based navigation, and smooth transitions. Technical implementation uses internal CSS, list-based navigation structure, hover animations with transform effects, and consistent iconography.

Footer Component (footer.html)

The footer component provides copyright and developer information with consistent visual design. It includes a left section with copyright notice and right section with developer credits. Features include 60px fixed height, gradient background, border top styling, minimalist design, and professional presentation. Technical implementation uses internal CSS, flexbox layout, consistent color scheme, and simple clean design.

Technical Architecture and Implementation

CSS Implementation Methods

The website implements all three CSS methods as required. External CSS (assets/css/styles.css) handles main theme, component styling, global styles, responsive design, color scheme, and typography. Internal CSS (style tags in components) handles header, navigation, and footer styling, component-specific styles, and frameset styling in index.html. Inline CSS (style attributes) provides page-specific enhancements, watch page styling, sign-up form styling, and responsive adjustments.

Design System

The design system uses deep black backgrounds (#0a0a0a, #1a1a1a) with vibrant green accents (#00ff88) for a sophisticated appearance. Typography uses Poppins font family with weights 300, 400, 500, 600, and 700. Components include consistent card designs, buttons, and form elements. Animations feature smooth transitions, hover effects, and shimmer effects.

Form Validation

The form validation system uses HTML5 native validation with no JavaScript dependencies. Features include pattern matching for usernames and passwords, visual feedback (green/red borders), real-time validation, and accessibility features (proper labels and form structure).

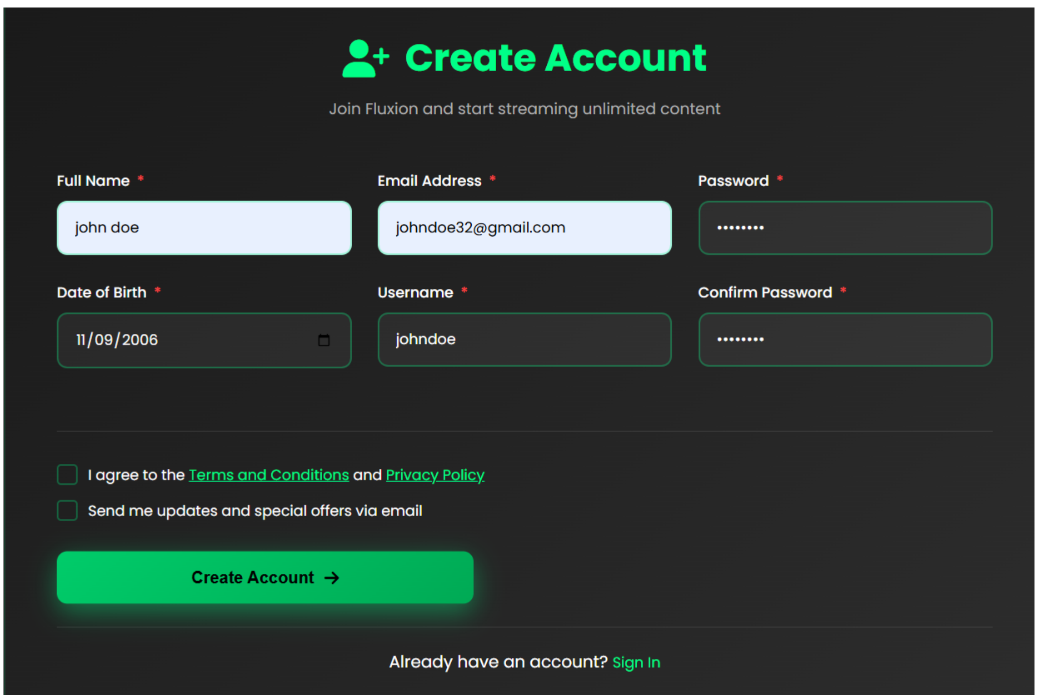
Conclusion

Fluxion is a movie streaming website that combines traditional HTML structure with modern design. It includes multiple pages, user forms, video players, and a dark, Netflix-style theme. The site shows strong skills in HTML5 and CSS3, with a responsive layout, clear navigation, and proper accessibility.

Overall, Fluxion proves that a professional and functional streaming platform can be built using clean, validated HTML and CSS balancing both educational goals and real-world usability.

FLUXION STREAMING PLATFORM – CONTINUED

Register Page (register.html)



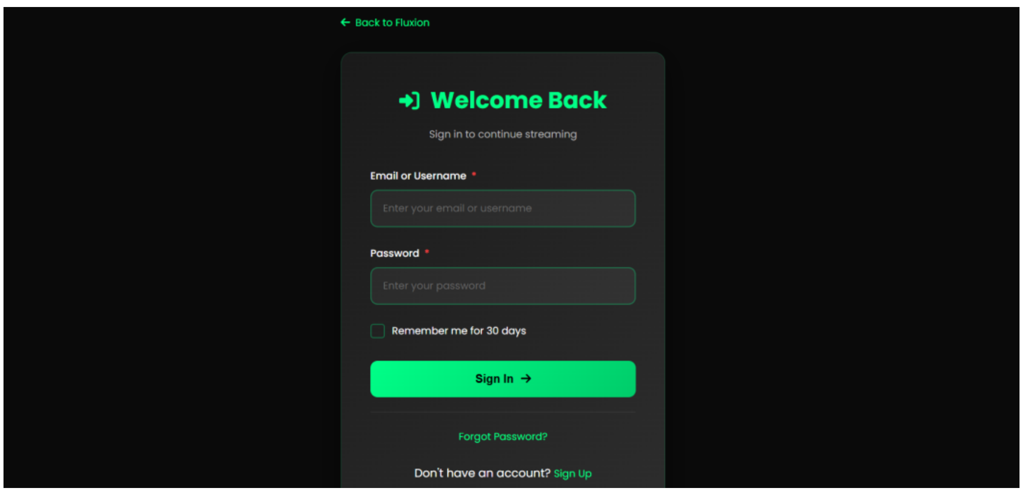
The register page provides the account creation system for new users within the website. It features a modern 3-column grid layout that organizes form fields into three structured sections: Personal Information (full name, date of birth), Account Details (email, username), and Security (password and confirm password). The page opens in a new browser tab and is designed for optimal clarity and accessibility.

Key features include real-time validation using jQuery to verify input values: email must be in a valid format and not previously registered, username must meet 4–20 alphanumeric-character requirements, and passwords must contain at least 8 characters with **a** visual strength indicator displaying weak/medium/strong. The confirmation password field must match to proceed.

When the Create Account button is clicked, the system performs complete client-side validation and, if all fields are correct, generates a unique timestamp-based user ID, stores the user data into local Storage under fluxion users, and automatically logs the user in by creating a session in fluxion user. A success notification appears, followed by automatic redirection to index.html, where the header updates to show the logged-in user's name and logout option.

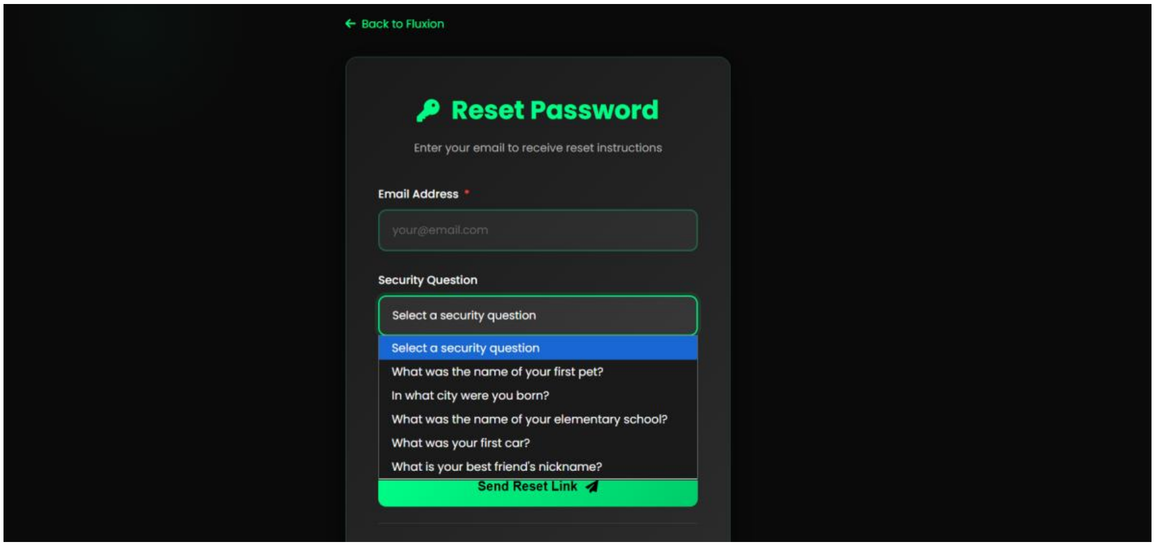
Technical implementation uses CSS Grid for responsive structure that adapts to a single-column layout on smaller screens, and jQuery event handling for real-time checking and validation feedback. Error messages appear below the respective input fields in red and include border colour changes for clarity, providing a smooth, professional, and user-friendly registration experience.

Login Page (login.html)



The login page is a centered, single column standalone page (maximum width 450px) that allows existing users to authenticate using either their email address or username along with their password, providing flexibility in the login method. When a user enters their credentials and clicks "Sign In", the page validates that both fields are filled, then retrieves all registered users from local Storage 'fluxion_users' array and searches for a matching user using JavaScript's find () method to check if the entered email OR username matches any user and the password matches exactly. If a match is found, the system removes the password from the user object for security purposes, saves the sanitized user object to 'fluxion_user' to create an active session, shows a success alert message, and redirects to index.html where the header automatically detects the session and updates to display the user's avatar (first initial), full name, and logout button. If no match is found, the system displays an error alert stating "Invalid email/username or password" without revealing which specific field was incorrect for security reasons. The page includes a "Remember Me" checkbox for future implementation, features a "Back to Fluxion" link for easy navigation, and uses jQuery for event handling and DOM manipulation combined with the Fluxion Auth class for centralized authentication logic, with set Time out to simulate processing delay for better user experience and all password fields properly masked using type="password" attribute.

Forgot Password Page (forgot-password.html)

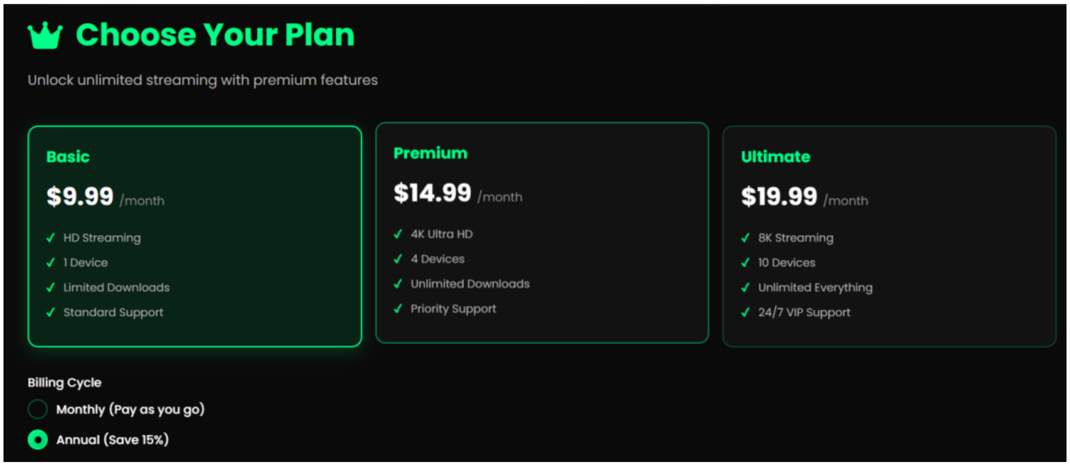


The forgot password page is a centered standalone interface (maximum width 480px) designed to guide users through a clear step-by-step password recovery process. Users begin by entering their registered email address in a validated input field, then choose a security question from a dropdown list containing options such as "What was the name of your first pet?", "What city were you born in?", and "What is your favourite book?". After selecting a question, they provide the corresponding answer in a text input field, which is verified in a case-insensitive manner.

When the user clicks the "Reset Password" button, the page validates that the email format is correct, a security question has been selected, and an answer has been provided. Any invalid input triggers specific inline error messages beneath the affected fields. In this demonstration version, the page simulates a successful submission to showcase the entire recovery flow. However, in a production environment, the system would verify that the email exists, check whether the provided security answer matches the stored one, and then either generate a temporary password or create a secure time-limited reset link, which would be sent to the user's registered email.

The page also includes helpful navigation links that allow users to return to the login page if they remember their password, or contact support for further assistance. Form handling uses HTML5 validation combined with jQuery, illustrating recommended practices for password recovery flows, while noting that a fully functional implementation would require backend integration with email services and secure token generation.

Purchase Page (purchase.html)



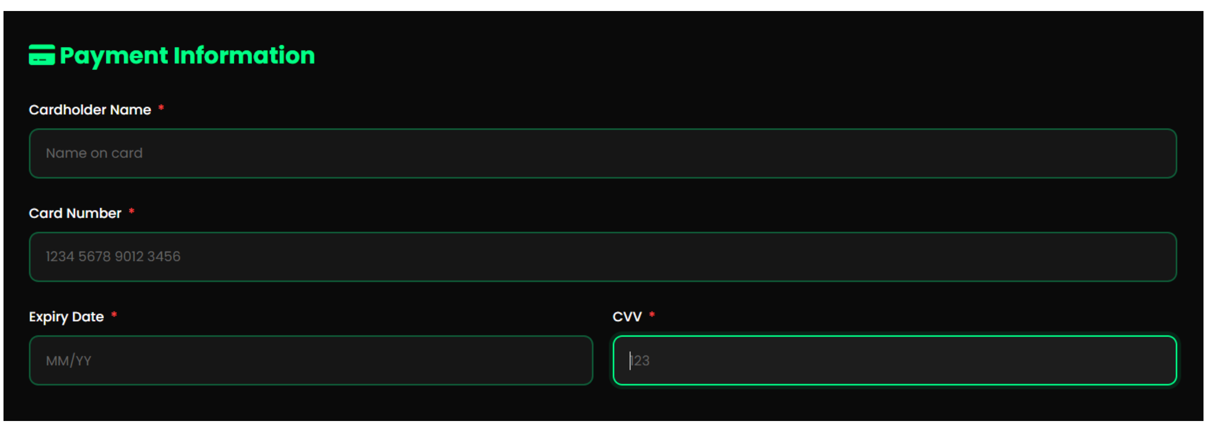
The purchase page provides a comprehensive e-commerce checkout experience within the main content frame, allowing users to subscribe to premium Fluxion plans through a complete payment flow including plan selection, billing options, and simulated payment processing.

The page displays three subscription tiers as interactive cards:

- Basic — $9.99/month, HD streaming, 1 device
- Premium — $14.99/month, 4K Ultra HD, 4 devices *(selected by default)*
- Ultimate — $19.99/month, 8K streaming, 10 devices

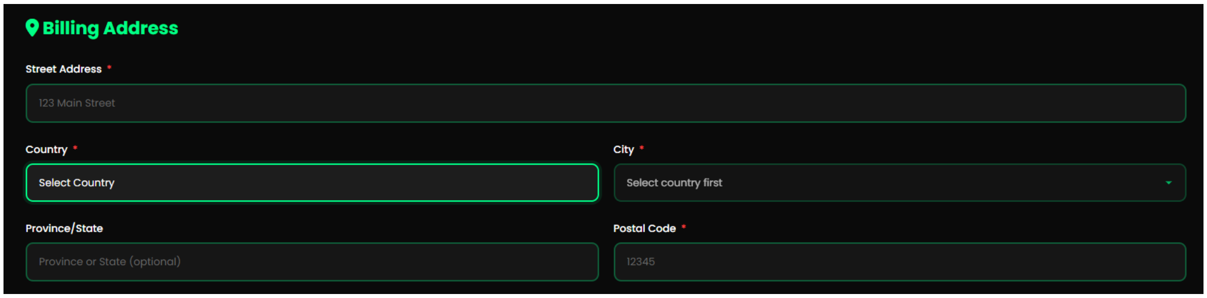
Users can then choose between Monthly billing (pay-as-you-go) or Annual billing (save 15% by paying for 12 months upfront). Selecting a billing cycle triggers real-time recalculation of the pricing summary, including subtotal, 10% tax, and final total.

Payment Information

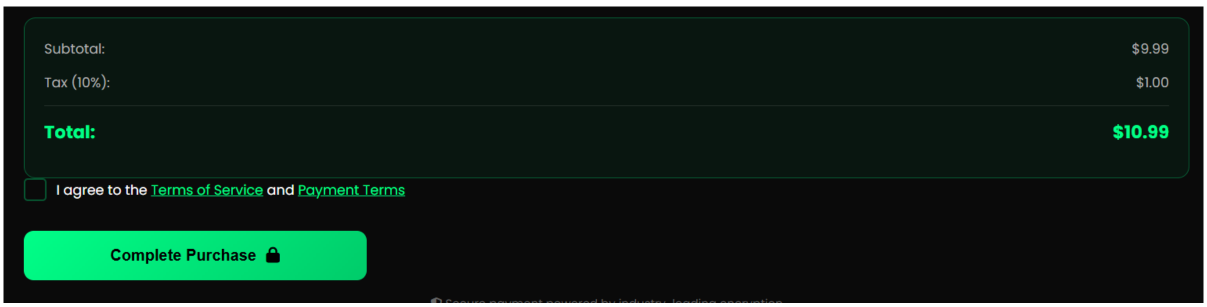


The payment information section includes: Cardholder name (minimum 3 characters), Card number, date with auto-formatting (MM/YY), CVV (3–4 digits), All fields are validated using jQuery and regex patterns.

Billing Address

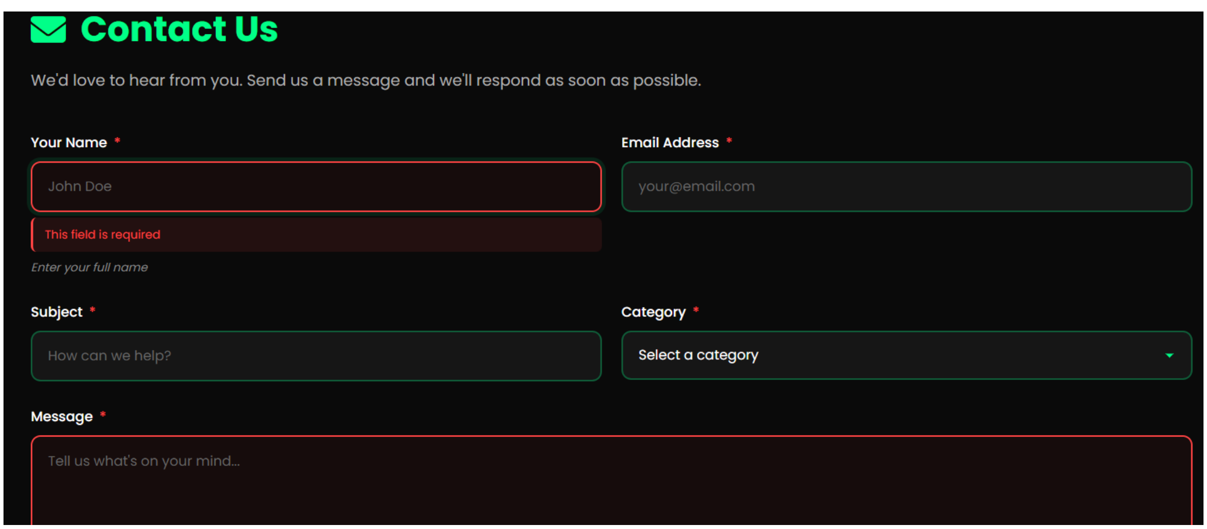


The billing address section collects the user's required location and contact information before completing the purchase. It includes fields for Street Address, Country, City, Province, Postal Code.



Below the address fields, the page displays a real-time Order Summary that updates based on the selected plan and billing cycle, The total reflects the Basic plan's monthly price ($9.99) plus a 10% tax applied automatically.

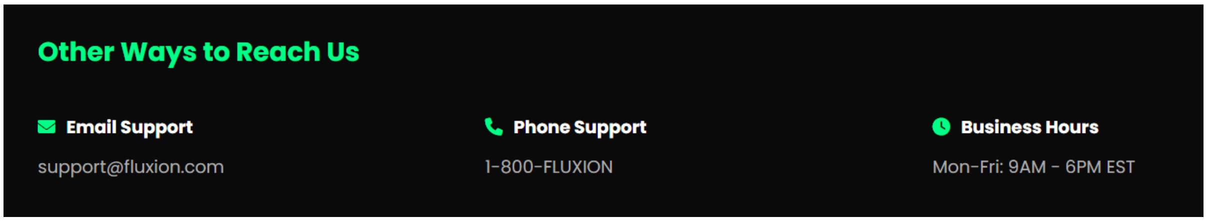
Contact Page (contact.html)



The Contact Us page provides users with a support form to send inquiries, feedback, or requests directly to the Fluxion team. The layout uses a structured two-column format, beginning with fields for Full Name and Email Address, followed by Subject and a Category dropdown that includes options such as General Inquiry, Technical Support, Billing & Payments, Content Request, Feedback, and Other.

A full-width Message text area allows users to describe their concern in detail, with a built-in character counter showing the remaining limit out of 500 characters. The page also provides an optional "Send me a copy of this message" checkbox that enables users to receive a copy of their submission.

When the "Send Message" button is clicked, validation ensures all required fields are completed correctly, such as entering a valid email and providing a message. If any required information is missing, inline red error messages appear beneath the affected fields. When the form is successfully validated, a simulated submission flow is displayed to represent a completed request process.



The page also displays contact details such as support@fluxion.com, a placeholder phone number, and support hours (Mon–Fri 9AM–6PM EST, Weekend 10AM–4PM EST). The entire implementation uses jQuery for validation and submission, CSS Grid for the responsive layout (stacking to one column on mobile), a character-counting function for the message box, filename preview for attachments, and automatic form reset after successful submission.

Summary

These five new pages transform Fluxion from a static demonstration into a functional web application with complete user management (register, login, forgot password), commerce capabilities (subscription purchase with payment processing), and customer support systems (contact form), all implemented using client-side technologies including HTML5, CSS3, jQuery, and local Storage API. The pages demonstrate industry-standard patterns for user authentication flows, form validation with real-time feedback, e-commerce checkout processes, and support ticket submission, while maintaining Fluxion's modern dark theme design with green accents throughout. Although designed for educational purposes with client-side storage and simulated backend processes, the implementation showcases professional development practices including comprehensive input validation, responsive layouts using CSS Grid and Flexbox, accessibility considerations with semantic HTML and proper form labels, user experience enhancements like automatic formatting and visual feedback, and data persistence strategies using local Storage with structured JSON objects for users, sessions, and purchases.